Push 2 / Nuendo Bridge

User Guide & Installation Manual — Version 1.0.4

Compatible with Nuendo 14+ and Cubase 14+ — macOS and Windows

This guide covers installation, configuration, and use of the Push 2 / Nuendo Bridge.

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1. System Requirements

- Ableton Push 2 (USB), Steinberg Nuendo 14+ or Cubase 14+, macOS 11+ or Windows 10/11
- **Python 3.11.9 maximum** (3.9–3.11.9). Newer 3.12/3.13 builds are not yet supported by the audio/MIDI dependencies.
- libusb (macOS: `brew install libusb`)
- push2-python must be installed from source:
  `pip install git+https://github.com/ffont/push2-python.git`
- Plugin Mapper (optional): pip install fastapi uvicorn pedalboard

  Nuendo 15+ features: *Plugin Browser, DirectAccess insert control and the Channel Strip extended-parameter / EQ-curve pages require API 1.3. All other features work with Nuendo 14+.*

2. Installation — macOS

Option A: Standalone App (Recommended)

- Step 1: Install Homebrew (if not already installed) — open Terminal and run:
  `/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"`
  See https://brew.sh for details.
- Step 2: Install libusb — `brew install libusb`
- Step 3: Copy Push2 Nuendo Bridge.app to /Applications
- Step 4: The app is **not code-signed**, so macOS Gatekeeper blocks it on first launch. Open Terminal and remove the quarantine flag (match the version in the .app name to your copy):

```
xattr -dr com.apple.quarantine "/Applications/Push2 Nuendo Bridge_v1.0.4.app"
```

- Step 5: Copy Ableton_Push2.js to:
  ~/Documents/Steinberg/Nuendo/MIDI Remote/Driver Scripts/Local/Ableton/Push2/
  Create the Ableton/Push2 folder if it doesn't exist.
- Step 6: Double-click the app. A P2 icon appears in the menu bar.

3. Installation — Windows

Windows now ships as a **standalone .exe** — no Python, pip or command line required. See the separate Windows Installation Guide for the step-by-step walkthrough. Key points:

- Download **Push2NuendoBridge-vX.Y.Z-Windows.zip** and unzip it anywhere (e.g. Documents). It contains the `.exe`, `Ableton_Push2.js` and the PDF guides.
- Install **loopMIDI** and create the four ports: `NuendoBridge In`, `NuendoBridge Out`, `BridgeNotes`, `BridgeNotes In`. Set it to start with Windows.
- Install the **WinUSB driver for the Push 2 with Zadig** (one time) so the bridge can talk to the device over USB. **Critical:** target the Push 2's **display/bulk interface** (USB ID `2982:1967`) — **NOT** the MIDI interface, or you will break MIDI on the Push. Full step-by-step in the Windows Installation Guide.

__IMPORTANT:__ installing the WinUSB driver with Zadig __disables the Push 2 in Ableton Live__ on this PC — Windows allows only one driver per device, and Live needs Ableton's own driver. The bridge and Ableton Live cannot use the Push 2 at the same time. This is fully reversible: see “Reverting the Zadig driver” in the Windows Installation Guide to restore Push 2 use in Live. (macOS is not affected.)

- **Double-click the .exe** to run the bridge — a console window shows the status and a clickable Plugin Mapper link (http://localhost:8100).

  No Python install is needed: the interpreter, all dependencies and the libusb runtime are bundled inside the .exe.

- Copy **Ableton_Push2.js** to the MIDI Remote scripts folder, creating the sub-folders if needed:

```
Nuendo:
C:\Users\<user>\Documents\Steinberg\Nuendo\MIDI Remote\Driver
Scripts\Local\Ableton\Push2\
```

```
Cubase:
C:\Users\<user>\Documents\Steinberg\Cubase\MIDI Remote\Driver
Scripts\Local\Ableton\Push2\
```

  Replace <user> with your Windows user name. The `Local\Ableton\Push2` *folders usually do not exist yet — create them.*

4. First Launch

Connect Push 2 via USB, launch the bridge, open Nuendo. Menu bar icon: P2 ✓ = connected, P2 ■ = waiting. Click for Start/Stop, Show Logs, Plugin Mapper, Start at Login, Quit.

5. Nuendo Configuration

Automatic setup: Nuendo usually detects the script and assigns its ports on its own.

If the script does not load automatically

- Open any project.
- Studio → MIDI Remote Manager → **Add MIDI Controller Surface**.
- Vendor = **Ableton**, Model = **Push2**.
- MIDI Input = **NuendoBridge Out**, MIDI Output = **NuendoBridge In** (the ports carry the same names as listed here).

Note Input

Set the instrument track's MIDI input to **BridgeNotes** (enable it in MIDI Port Setup if hidden).

Important — avoid doubled notes

- In Studio Setup → MIDI Port Setup, **uncheck “In ‘All MIDI Inputs’” for both Push 2 ports** — “Ableton Push 2” (Live Port) and “Ableton Push 2 User Port”.

  If left checked, every pad you play is received twice: once via the bridge's BridgeNotes port and once directly from the Push 2 hardware. Excluding the raw Push 2 ports is standard practice for any control-surface bridge and is permanent once set.

6. Using the Bridge

8 vertical zones on the Push 2 screen. Left/Right = bank by 8 tracks. Shift+Left/Right = nudge by 1 track (Volume/Pan modes).

Button	Mode	Encoders
Mix	Volume	8 track volumes
Shift+Mix	Track	Vol+Pan+Sends
Clip	Sends	8 sends
Shift+Clip	Pan	8 track pans
Device	Quick Controls	8 QC
Browse	Inserts	Plugin params
Add Device	Plugin Browser	Browse/load plugins

Lower row: Mute toggles Mute/Monitor. Solo toggles Solo/Rec. Shift+Mute/Solo clears all.

7. Mixer Modes

Volume Mode (Mix button)

Shows 8 tracks with volume bars, VU meters, and peak clip indicators. Turn an encoder to adjust volume.

Upper row buttons (above the display):

- Short press = select track
- Shift + press = clear peak clip for that track
- Long press (1 second) = open instrument UI
- Double press = open Edit Channel Settings

Lower row buttons (below the display):

- Toggle Mute, Solo, Monitor, or Record Arm on the corresponding track
- Press the Mute button to switch between Mute and Monitor mode
- Press the Solo button to switch between Solo and Record Arm mode
- Long press (0.5s) in Mute mode = toggle Monitor on that track
- Long press (0.5s) in Solo mode = toggle Record Arm on that track
- Shift+Mute = clear all mutes (or all monitors in Monitor mode)
- Shift+Solo = clear all solos (or all rec arms in Rec mode)

Pan Mode (Shift+Clip)

Shows 8 tracks with pan position indicators. Turn an encoder to adjust pan.

Track Mode (Shift+Mix)

Combined mode for the selected track: Enc1=Volume, Enc2=Pan, Enc3-8=Sends 1-6.

8. Sends Mode

Press Clip. Encoders = send levels. Upper row = On/Off. Lower row = Pre/Post. Left/Right = navigate tracks.

9. Inserts Mode

List View: Short press upper = params. Long press = open UI. Shift+upper = params+UI. Lower = bypass. Shift+Left/Right = slots 1-8 / 9-16.

Parameter View: Encoders = 8 params. Left/Right = bank/page nav. Lower 1=UI, 2=bypass, 3=deactivate. If mapped: ★ MAPPED indicator.

10. Plugin Browser

Requires Nuendo 15+ (MIDI Remote API 1.3)

Press Add Device to open the browser.

Phase 1 — Slot Selection

The screen shows the insert slots of the selected track.

- Upper row buttons (above the display) = select the target slot
- Left/Right = switch between slots 1-8 and 9-16
- Lower row button 8 = Cancel

Phase 2 — Plugin List

- Encoder 1 = page scroll (8 at a time)
- Encoder 2 = fine scroll (highlight moves within page)
- Upper row = load plugin from that column
- Lower row button 1 = cycle to next collection
- Lower row button 8 / Browse = back / cancel

After loading, the bridge returns to Inserts mode and rescans slots automatically.

11. Quick Controls (Device Mode)

Press Device. 8 Quick Controls of the selected track. Left/Right = navigate tracks.

12. Transport Controls

Button	Function	LED
Play	Playback	White/Green/Purple
Record	Recording	Orange/Blink Red/Solid Red
Fixed Length	Cycle/Loop	
Automate	Automation cycle	White/Green/Red
Metronome	Metronome	LED reflects state

13. Automation

Automate button cycles: Off → Read → Read+Write → Write → Off.

14. Touchstrip

Shift+Layout to cycle: Pitch Bend, Mod Wheel (CC 1), Volume.

15. Note Input

64 pads: Chromatic (default) or Drum (Layout). Scale, Accent, Note Repeat available. Set instrument input to BridgeNotes.

Scale selector (Scale button): pick a musical scale or the **Piano** layout. Piano mode arranges the pads like a piano keyboard — white keys on the lower row of each octave pair, black keys on the upper row, four octaves stacked bottom-to-top, every C in purple. The root is locked to C in Piano mode; use Octave Up/Down to move the whole four-octave block.

16. Control Room

Press User. 4 pages: Main, Phones, Cues, Sources. Master encoder = Main level. User+Master = Phones.

17. Channel Strip

> *Extended-parameter and EQ-curve control require Nuendo 15+ (API 1.3).*

From the Mixer view, press **Mix again** to enter Channel Strip mode. The overview shows the six strip sections as Cubase-coloured banners (Gate, Comp, EQ, Tools, Sat, Limiter) plus PreFilter Phase and PreGain.

> ***Important:** a strip module can only be controlled once it has been **activated manually in Nuendo**. Open the channel's strip in the Nuendo mixer and load/enable each module (Gate, Comp, EQ, Tools, Sat, Limiter) you want to use. An empty or disabled strip slot is not exposed to the Push 2 — its banner and encoders stay inactive until you turn the module on in Nuendo.*

Overview

- Each banner shows the section colour (dimmed when bypassed) and the loaded plugin variant.
- Lower row 1-6 = Bypass each section (pill/LED amber when bypassed).
- Lower row 7 = PreFilter Phase On/Off. Lower row 8 = Edit Channel Settings (white when the window is open).
- Upper row 1-6 = drill into that module's sub-page.

Module Sub-pages

- 8 encoders control that module's parameters. Parameters beyond Cubase's bank zone are exposed via DirectAccess (e.g. VintageCompressor Ratio & Mix, Tube Compressor Character, DeEsser side-chain filter, Maximizer Release/Recover).
- Lower row = the module's binary toggles (Auto Release, Solo, Diff, etc.) — LEDs follow the on-screen pills.
- Upper row layout: 1 = module name (inactive), 2 = Bypass, 3-5 = variant chips, 8 = back to overview.

  Variant chips show the currently loaded plugin. Switching the variant must be done in Nuendo's strip-slot menu — the MIDI Remote API does not allow changing strip-slot plugins from a controller; the chip mirrors the active variant in real time.

18. Channel EQ Page

Drill into the EQ section for an EQ-Eight-style page with a live frequency-response curve.

Encoder	Function
1	Band selector (1-4) — turn to pick the active band
2	Filter Type of the selected band (bands 2-3 are fixed Peak)
3	Frequency of the selected band
4	Q of the selected band
5	Gain of the selected band
6	PreFilter Low-Cut frequency

7	PreFilter High-Cut frequency
8	PreGain

- Lower row 1 = selected band On/Off. Lower 6 = LC On, Lower 7 = HC On, Lower 8 = PreFilter bypass.
- Upper row 2 = EQ section bypass (amber when bypassed). Upper 8 = back.
- The curve combines the 4 EQ bands + PreFilter HC/LC. Numbered circles mark each band at its (freq, gain) point — white when the band is on, grey when off. The curve and markers update live, whether you change a parameter from the Push or directly in Nuendo.

19. Setup Page

Press Setup. Tabs: MIDI Ctrl (aftertouch), Vel Curve (5 presets), CC Mode (Absolute/Pick-up), About.

CC Mode

- **Absolute** (default): encoder value is sent immediately. May cause parameter jumps if the encoder position differs from the current value in Nuendo.
- **Pick-up**: encoder does not send values until the user reverses the direction of rotation. This prevents jumps but requires a direction change to engage.

  Important: *Nuendo does not send the current CC parameter value back to the Push 2. Because of this limitation, the bridge cannot know the actual value in Nuendo and cannot implement a true pick-up (where the encoder catches up to the existing value). Instead, pick-up mode uses direction-change detection: turn the encoder in one direction (nothing happens in Nuendo), then reverse — the encoder engages from that point on.*

20. MIDI CC Controller

Shift+Note. 8 assignable CC faders on BridgeNotes port. Upper row = edit CC number. Lower row = toggle 0/127. Defaults: CC 1,2,7,8,10,11,64,65.

21. Plugin Mapper

Create custom parameter mappings for VST3 plugins. Access via menu bar (Plugin Mapper) or http://localhost:8100. Mappings saved in **~/.push2bridge/mappings/**. See the separate Plugin Mapper Guide for details.

Requires: pip install fastapi uvicorn pedalboard. The bridge works without these — the Mapper is optional.

22. Button Reference

Button	Action	Shift + Button
Mix	Volume mode (press again: Channel Strip)	Track mode
Clip	Sends mode	Pan mode
Device	Quick Controls	
Browse	Inserts mode	
Add Device	Plugin Browser	
Note	MIDI note pads	MIDI CC controller
Setup	Setup page	
Left/Right	Bank nav (8)	Nudge (1) in Vol/Pan
Play	Playback	

Record	Record toggle	
Mute	Mute/Monitor	Clear all
Solo	Solo/Rec arm	Clear all
User	Control Room	
Layout	Drum/Chromatic	Touchstrip cycle
Upper (dbl / Shift)	Edit Channel Settings	
Scale	Scale selector	
Accent	Fixed velocity	
Repeat	Auto-repeat	

23. MIDI Channel Allocation

Channel	Usage
1 (0xB0)	Mixer (volume, pan, transport, selection, VU)
2 (0xB1)	Insert plugin parameters
3 (0xB2)	Send levels (selected track)
4 (0xB3)	Insert bypass toggles
5 (0xB4)	Quick Controls
6 (0xB5)	Control Room
7 (0xB6)	Bank zone sends
8 (0xB7)	DirectAccess commands (browser, bypass, edit, strip slot, param-flip, variant)
9 (0xB8)	DirectAccess encoder control (mapped params, strip slots)
16 (0xBF)	Heartbeat

24. Troubleshooting

- **Push 2 not found**: check USB, libusb, close Ableton Live.
- **push2-python error**: install it from source — pip install git+https://github.com/ffont/push2-python.git. Use Python 3.11.9 or lower.
- **No Nuendo connection**: check bridge is running; add the surface manually (MIDI Remote Manager → Add, Vendor Ableton / Model Push2).
- **Pads play doubled notes**: uncheck “In ‘All MIDI Inputs’” for the Push 2 Live Port and User Port (Studio Setup → MIDI Port Setup).
- **Channel Strip / EQ curve not working**: requires Nuendo 15+ (API 1.3). Drill into a section once to trigger DA strip exploration.
- **Plugin Mapper not loading**: pip install fastapi uvicorn pedalboard.
- **Track names not loading**: navigate Left/Right to refresh.

  Logs: macOS — ~/Library/Logs/Push2NuendoBridge.log. Windows — terminal output.

25. Support

Push 2 / Nuendo Bridge is donationware.

- GitHub: https://github.com/mbourque-mix/Push2Nuendo-Bridge
- Buy Me A Coffee: https://buymeacoffee.com/mbourque
- Issues: https://github.com/mbourque-mix/Push2Nuendo-Bridge/issues

